Management Information Systems

Week 5

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Last Week…

- Managers must understand ***organizational*** structures, processes, culture, and incentives to successfully build and use IS.

- IS can significantly reshape organizational processes, decision-making, and performance, often driving organizational change.

- Strategic frameworks such as ***Porter's competitive forces, the value chain, core competencies, synergies, and network economics*** help firms use IS to gain competitive advantage.

- Strategic information systems pose challenges such as imitation, rapid technological change, and misalignment, which require continuous adaptation and strong managerial oversight.

Weekly Topics

- **Subject to change without prior notice**

W	Ch	Topic	W	Ch	Topic
1		Orientation Introduction to MIS	9	5	IT infrastructure and emerging technologies
2	1	Information Systems in Global business today	10	6	Foundations of Business Intelligence
3	2	Global E-business and Collaboration	11		*No Class*
4	3	Information Systems, Organizations, and Strategy	12	11	Managing knowledge
5	**9**	**Achieving operational excellence and customer intimacy**	13	12	Enhancing decision making
6	4	Ethical and Social issues in information systems	14	13	Building information systems Hands-on Session
7	7	Telecommunications, the internet, and wireless technology*	15	8	Securing information systems*
8		*Midterm Exam*	*16*		*Final Exam*

Chapter 9.

Achieving operational excellence and Customer Intimacy

Questions to Answer

1. How do ***enterprise systems*** help businesses achieve operational excellence?

2. How do ***supply chain management systems*** coordinate planning, production, and logistics with suppliers?

3. How do ***customer relationship management systems*** help firms achieve customer intimacy?

4. What are the ***challenges*** that enterprise applications pose, and how are enterprise applications taking advantage of ***new technologies***?

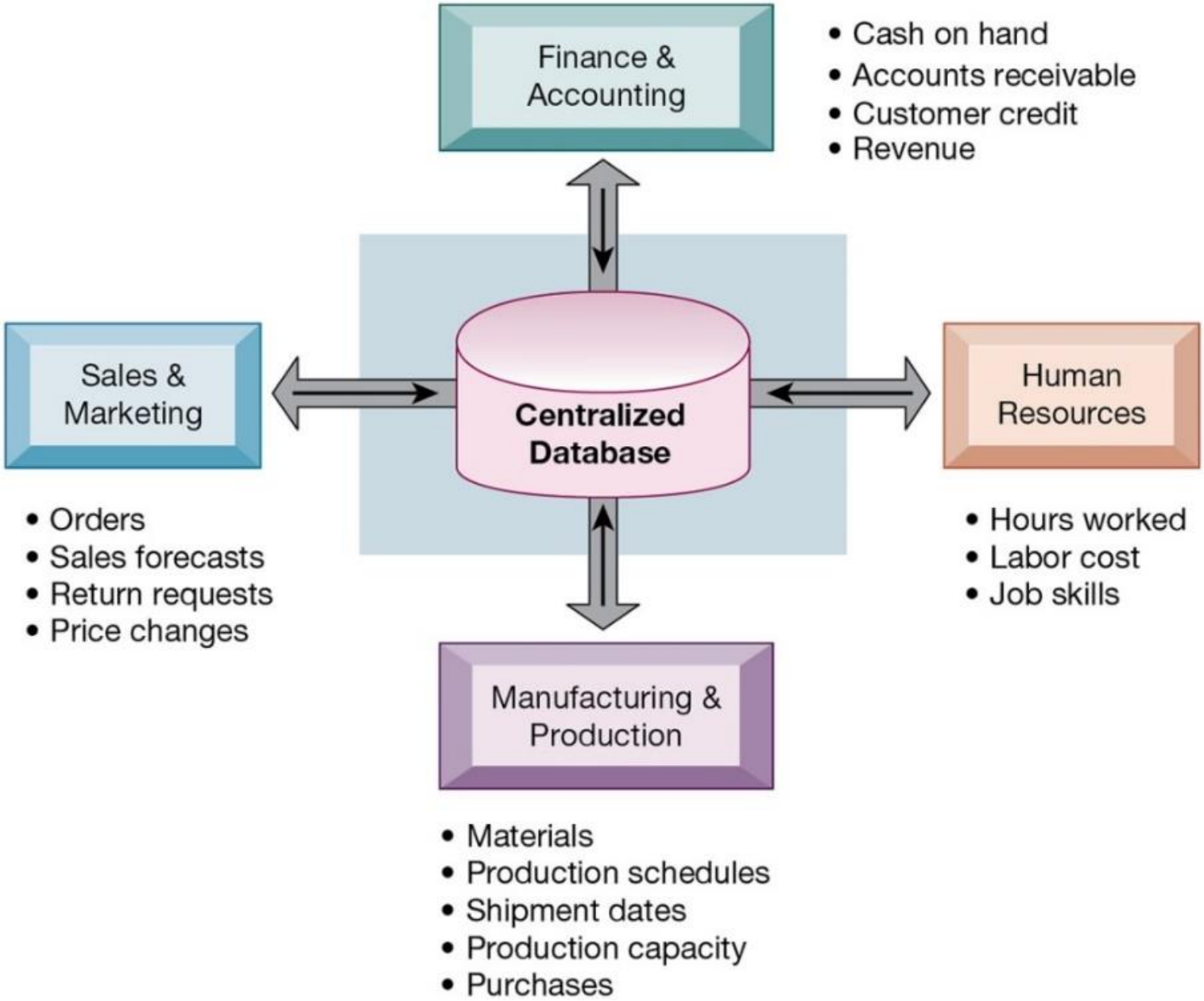
9.1
How do enterprise systems help businesses achieve operational excellence?

Enterprise Systems

- **Firms need a special enterprise system to *integrate* information**

- **Enterprise resource planning (ERP) systems**
  - Suite of integrated software ***modules*** and a common ***central*** database
  - Collects data from ***many divisions*** of firm for use in nearly all of firm's internal business activities
  - Makes the data available for applications that support nearly ***all*** of an organization's ***internal*** business activities.
  - Information entered in one process is immediately available for other processes

How Enterprise Systems Work



Enterprise Software

- **Built around thousands of predefined business processes that reflect best practices**
- **To implement, firms:**
  - Select functions of system they wish to use
  - Map business processes to software processes

[BUSINESS PROCESSES SUPPORTED BY ENTERPRISE SYSTEMS]

Financial and accounting processes, including general ledger, accounts payable, accounts receivable, fixed assets, cash management and forecasting, product-cost accounting, cost-center accounting, asset accounting, tax accounting, credit management, and financial reporting
Human resources processes, including personnel administration, time accounting, payroll, personnel planning and development, benefits accounting, applicant tracking, time management, compensation, workforce planning, performance management, and travel expense reporting
Manufacturing and production processes, including procurement, inventory management, purchasing, shipping, production planning, production scheduling, material requirements planning, quality control, distribution, transportation execution, and plant and equipment maintenance
Sales and marketing processes, including order processing, quotations, contracts, product configuration, pricing, billing, credit checking, incentive and commission management, and sales planning

Business Value of Enterprise Systems

- **Enterprise systems enhance operational efficiency and provide comprehensive information for better decisions.**
- **Large companies with multiple locations use enterprise systems to enforce *standard* practices and *data consistency* worldwide.**
  - e.g., Coca-Cola used SAP to standardize business processes across 200 countries, improving raw material purchasing and market responsiveness.
- **Enterprise systems integrate order, manufacturing, and delivery data, helping firms quickly *respond to customers*.**
  - Enable manufacturing to produce only what is ordered, procure the exact amount of raw materials, stage production efficiently, and minimize inventory time.

Business Value of Enterprise Systems (Cont.)

- **Enterprise systems provide valuable information for *management decisions* enabling more accurate sales and production forecasts.**
- **Enterprise systems include *analytical tools* to evaluate overall organizational performance**
  - Data captured is used for performance analysis and has common ***standardized definitions and format***s accepted across the organization.
  - Performance figures are consistent throughout the company.
- **Senior management can easily check the performance of any organizational unit.**
  - Identifies the most and least profitable products and calculates companywide costs, providing ***enhanced management insight***.

ERP Software

Figure 1: Magic Quadrant for Cloud ERP for Service-Centric Enterprises

Figure 1: Magic Quadrant for Cloud ERP for Product-Centric Enterprises

Source: Gartner

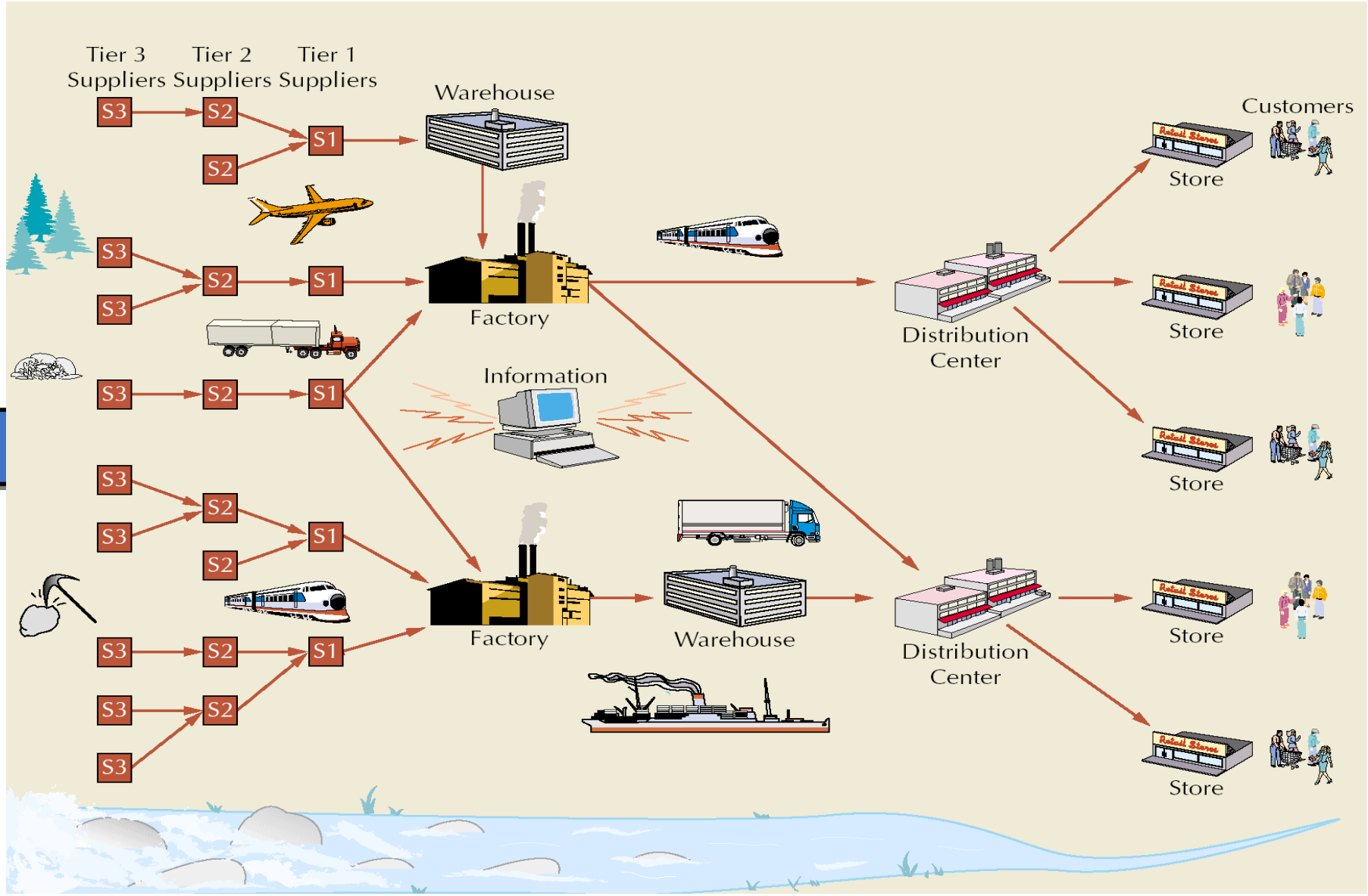
Video

- **SAP Tutorial for Beginners (12:40~)**

9.2

How do supply chain management systems coordinate planning, production, and logistics with suppliers?

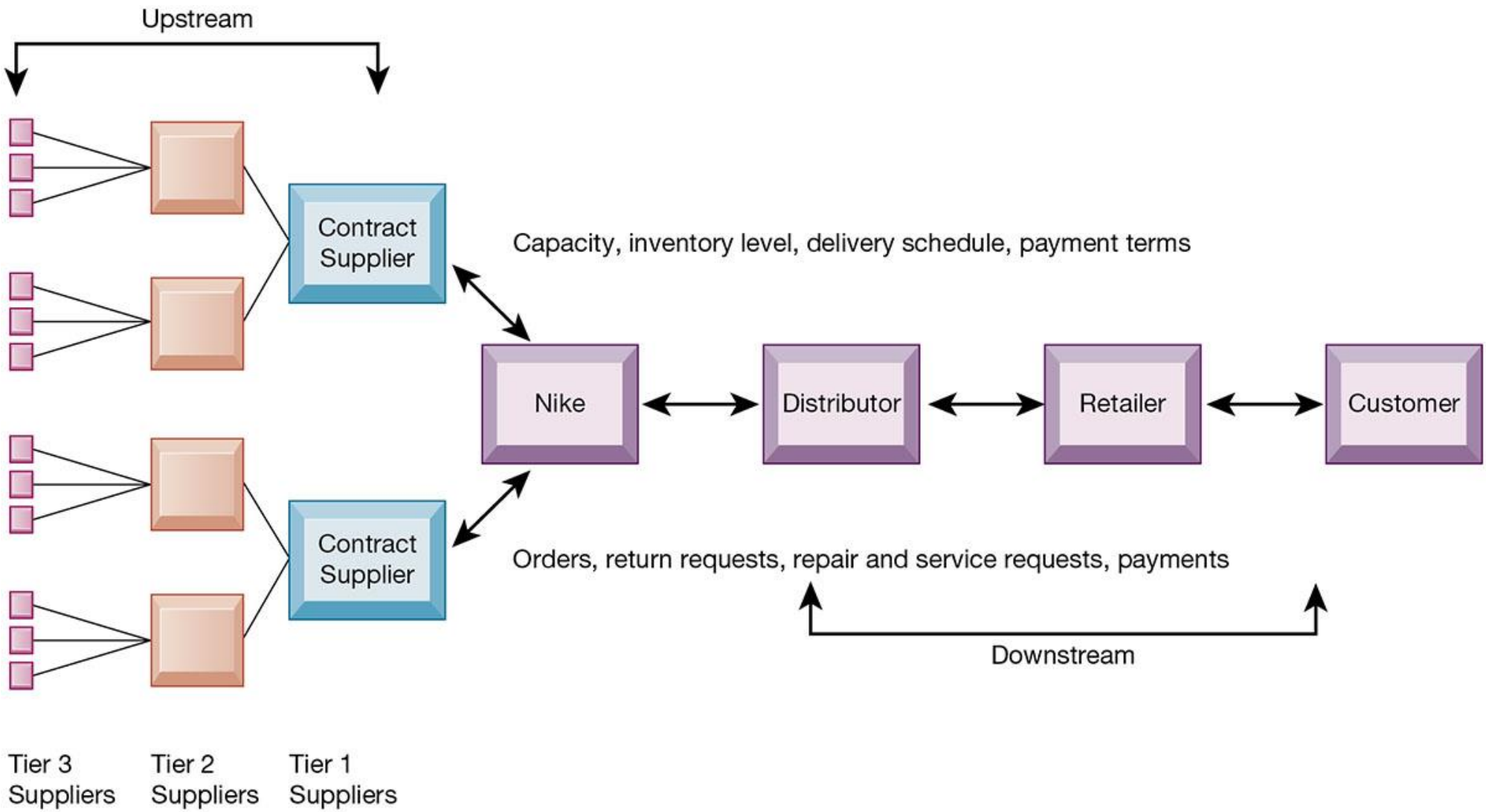
The Supply Chain



The Supply Chain

- **Network of organizations and processes for:**
  - ***Procuring*** materials, ***transforming*** them into products, and ***distributing*** them
  - SC ***links*** suppliers, manufacturing plants, distribution centers, retail outlets, and customers to supply goods and services from source through consumption.
  - Materials, information, and payments flow through the SC in ***both*** directions.
    - Raw materials – intermediate products (i.e., components or parts) – finished products – distribution centers – retailers – customers
- **Upstream supply chain**
  - Suppliers, suppliers' suppliers, processes for managing relationships with them
- **Downstream supply chain**
  - Organizations and processes responsible for delivering products to customers
- **Internal supply chain**

Nike's Supply Chain



Information Systems and SCM

- **Inefficiencies cut into a company's operating costs**
  - Inefficiencies in the SC are caused by ***inaccurate or untimely information***. (e.g., parts shortages, underused plant capacity)
  - Can waste up to 25 percent of operating expenses
- **Just-in-time strategy**
  - Need ***perfect information*** about exactly how many units of product customers want, when they want them, and when they can be produced
- **Safety stock: *Buffer* for lack of flexibility in the supply chain**
  - Many SC events are unpredictable, such as demand swings, late deliveries, defects, and breakdowns.
  - To handle this uncertainty, manufacturers often keep extra inventory ➔ costly, but low stock can also cause lost sales from canceled orders.

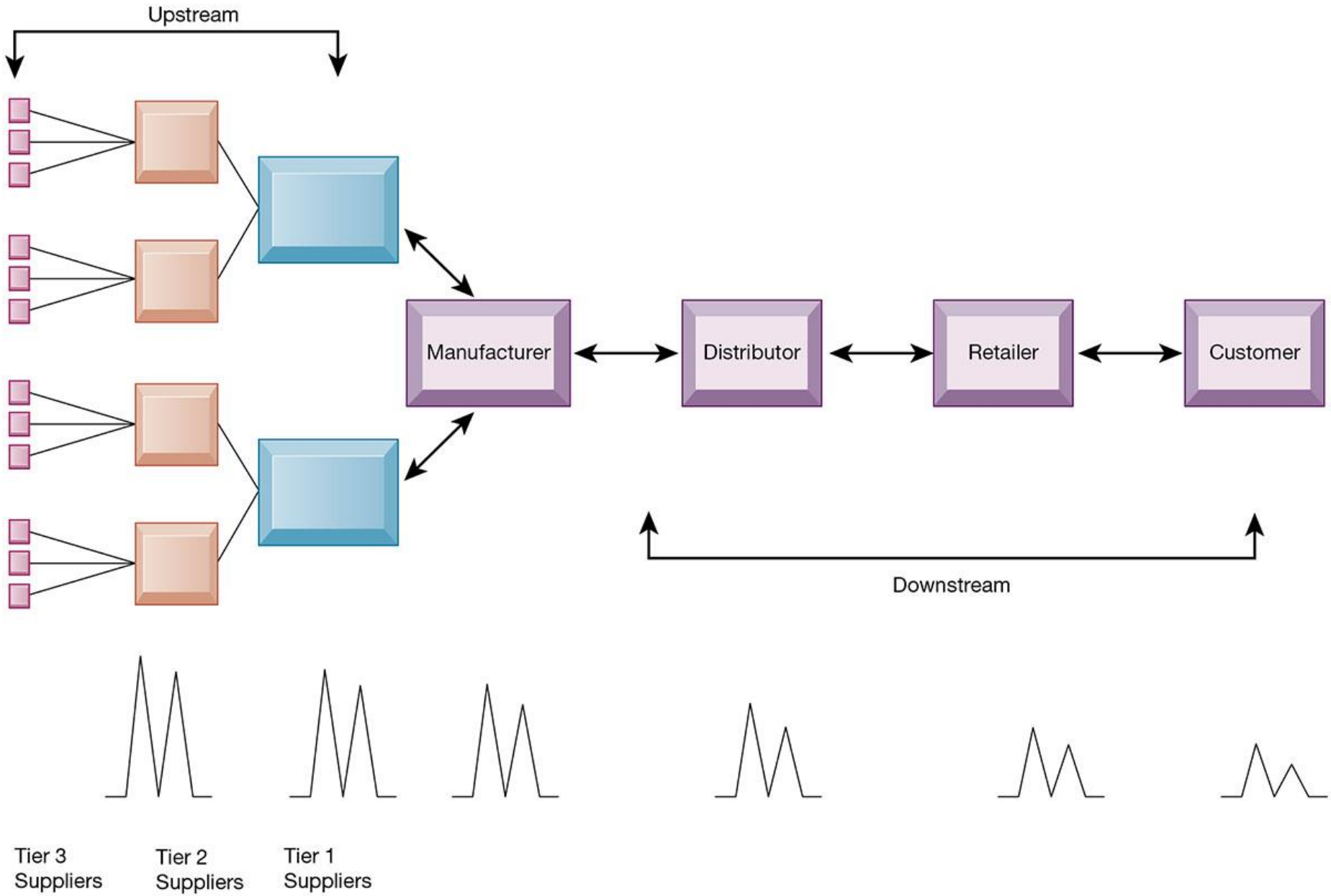
Bullwhip Effect

- **The bullwhip effect occurs when information about product demand *gets distorted* as it moves through the supply chain.**
- **Small changes in demand create *a ripple effect* throughout the supply chain.**

➔ excess inventory, increased production, warehousing, and shipping costs.

- **Solution: Reduce uncertainties about demand and supply by providing all SC members with accurate and latest information.**
  - ***Share dynamic information*** about inventory levels, schedules, forecasts, and shipments among all supply chain members.
  - Use SCM systems to help members make better purchasing and scheduling decisions.

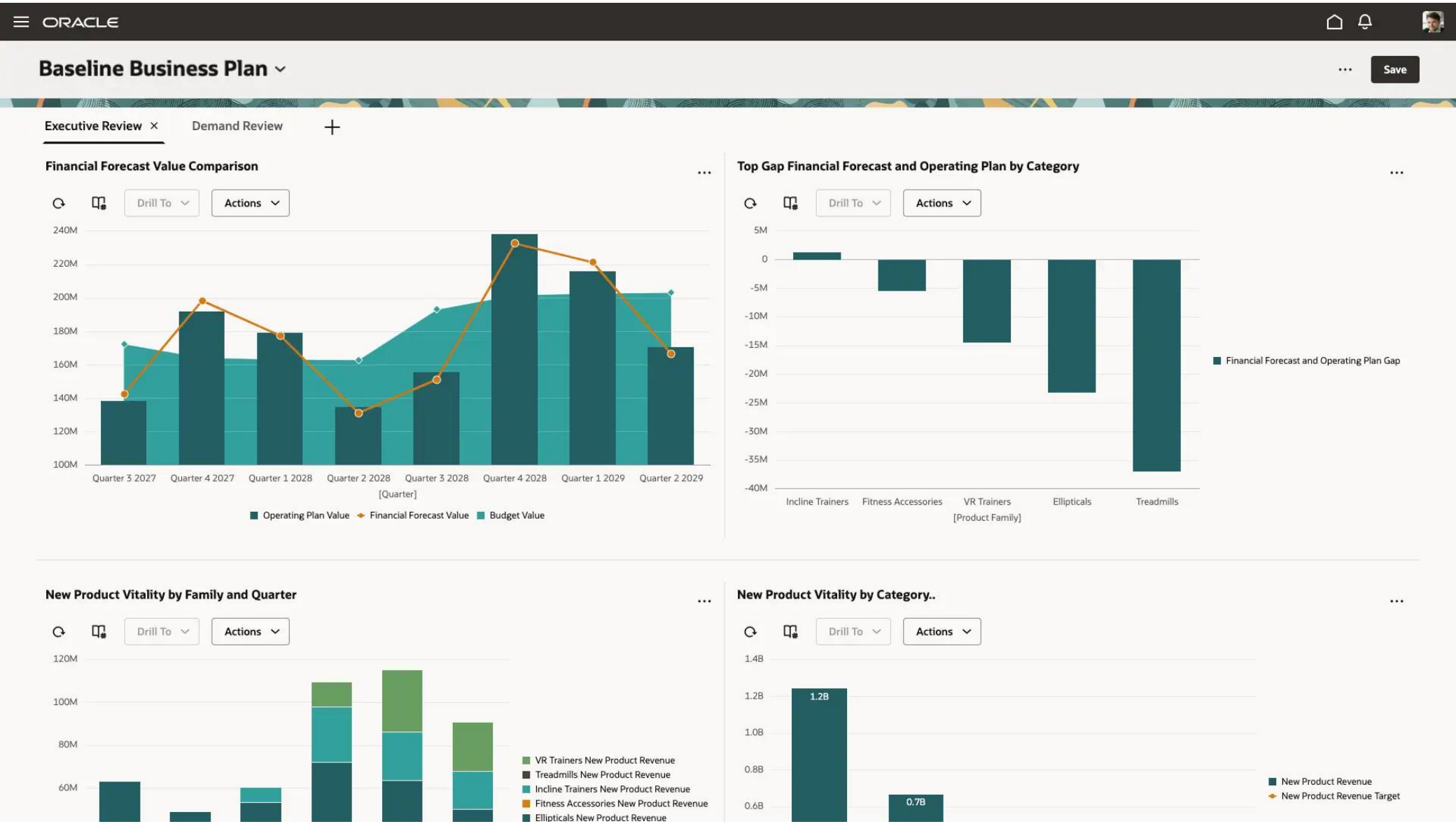
The Bullwhip Effect



Supply Chain Management Software for *Planning*

- **Supply chain software:**
  1) supply chain planning and **2) supply chain execution**

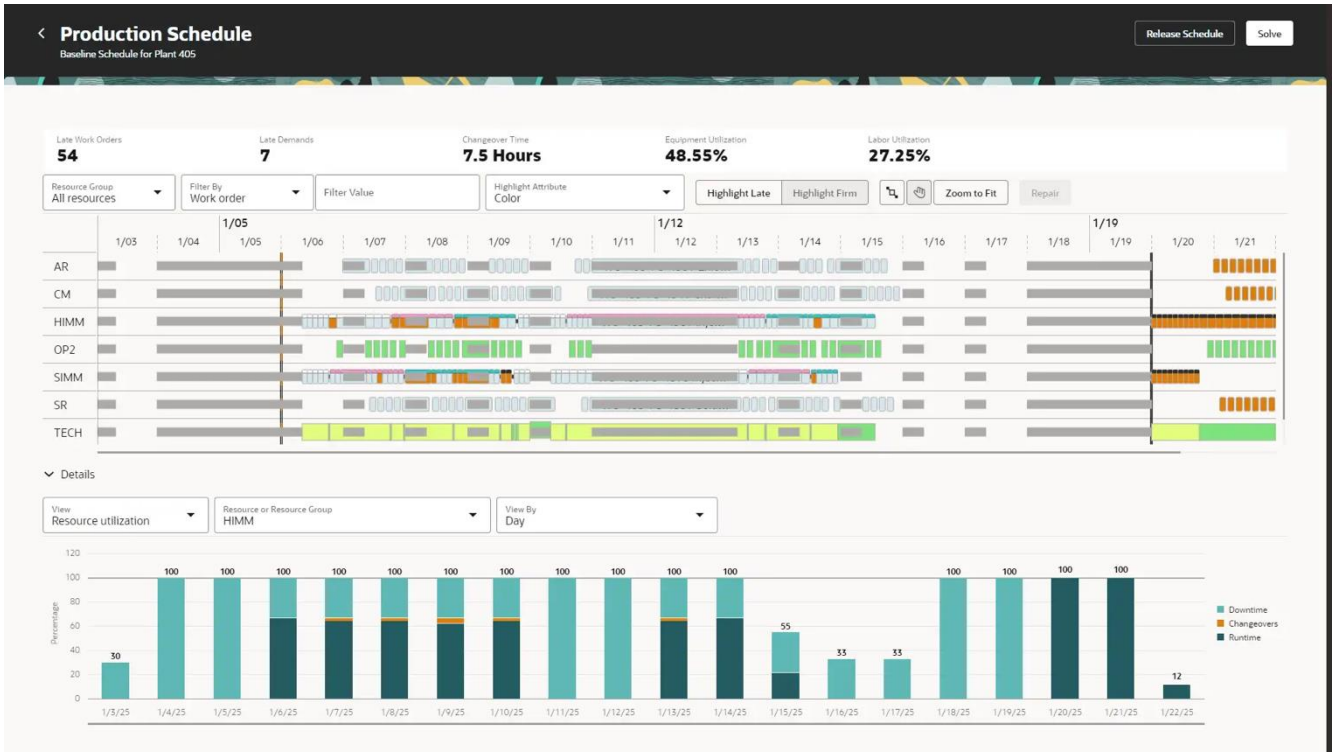
- **Supply chain *planning* systems**
  - Enable the firm to model its existing SC, generate demand forecasts for products, and develop optimal sourcing and manufacturing ***plans***.
  - ***Support better decision-making***: determining how much of a specific product to manufacture in a given time period; establishing inventory levels for raw materials, intermediate products, and finished goods; determining where to store finished goods; and identifying the transportation mode for delivery.
  - ***Demand Planning***: One of the most important and complex functions. Determines how much product a business needs to make to meet customer demand.



https://www.oracle.com/kr/scm/supply-chain-planning/

Supply Chain Management Software for *Execution*

- **Supply chain *execution* systems**
  - Manage the flow of products through distribution centers and warehouses to ensure that products are ***delivered*** to the right locations most efficiently
  - Track the physical ***status*** of goods, the management of materials, warehouse and transportation operations, and financial information involving all parties.



https://www.oracle.com/kr/scm/supply-chain-planning/

Supply Chains Before the Internet

- ***Internal*** **Coordination Problems:**
  - Difficult to make information flow smoothly ***among systems*** for purchasing, materials management, manufacturing, and distribution.

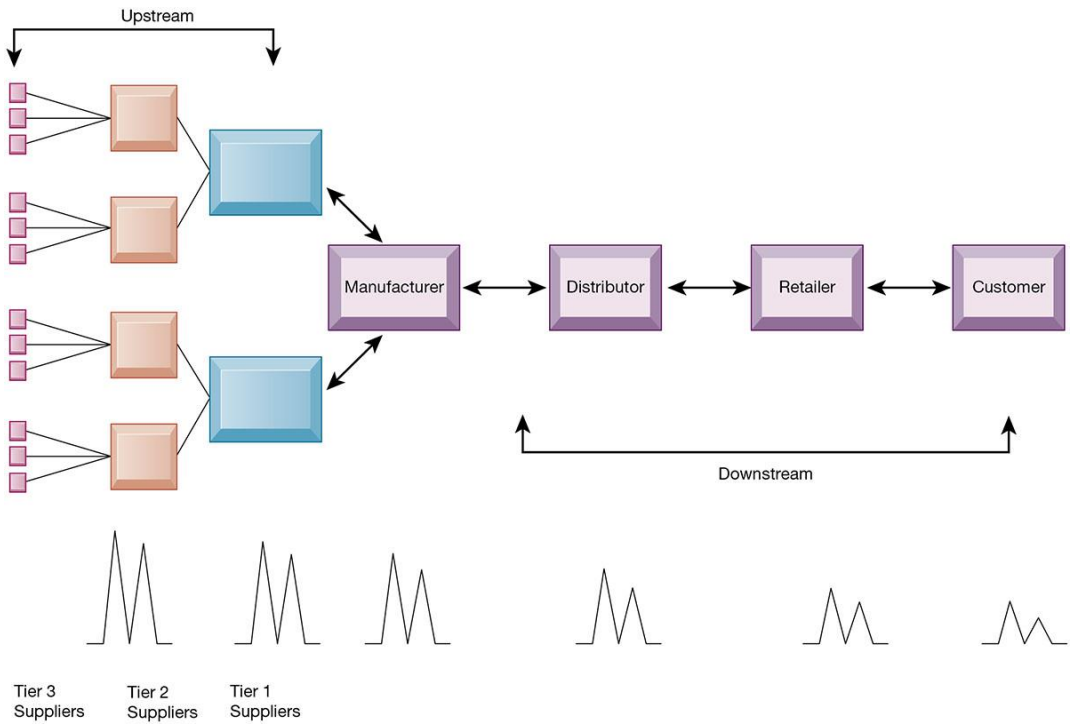
- ***External*** **Coordination Problems:**
  - ***Hard to share*** information with suppliers, distributors, and logistics providers due to incompatible technology platforms and standards.

Supply Chains after Internet Integration

- **Enterprise and SCM systems with Internet technology help *integrate* information flows.**
- **Managers use *web interfaces* to check if suppliers' inventory and production match demand.**
- **Business partners collaborate online using *web-based tools* for forecasting.**
- **Sales representatives *monitor* customers' order status via suppliers' production schedules and logistics information.**

Discussion

- **When a military conflict suddenly breaks out, governments may rush to order more weapons, ammunition, and equipment. What problems could this create in the defense supply chain? What is different about the weapons supply chain compared with the consumer product supply chain?**



Global Supply Chain Issues

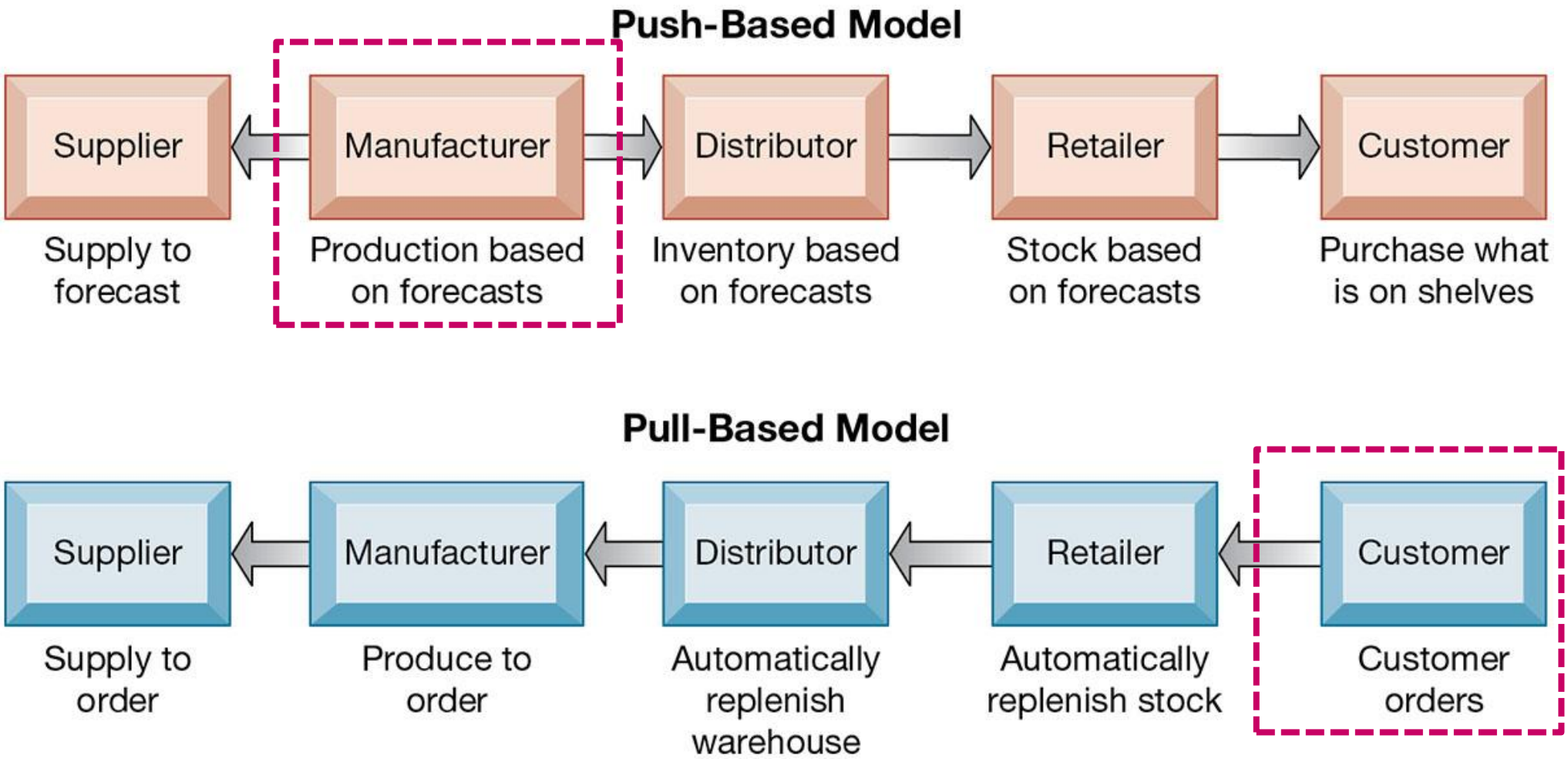
- **Companies' SCs extend across multiple countries and regions.**
  - Outsourcing manufacturing operations, obtaining supplies from other countries, selling abroad etc. ➔ additional complexities and challenges
- **Internet helps manage global complexities in sourcing, transportation, communications, and international finance.**
- **Firms relies heavily on *outsourcing* and *third-party logistics providers***
  - Outsourcing of warehouse management, transportation management, and related operations.
  - Third-Party Providers offer web-based software for better visibility into global supply chains.
  - Customers monitor inventory and shipments via secure websites, improving efficiency in managing global supply chains.

Demand-Driven Supply Chains

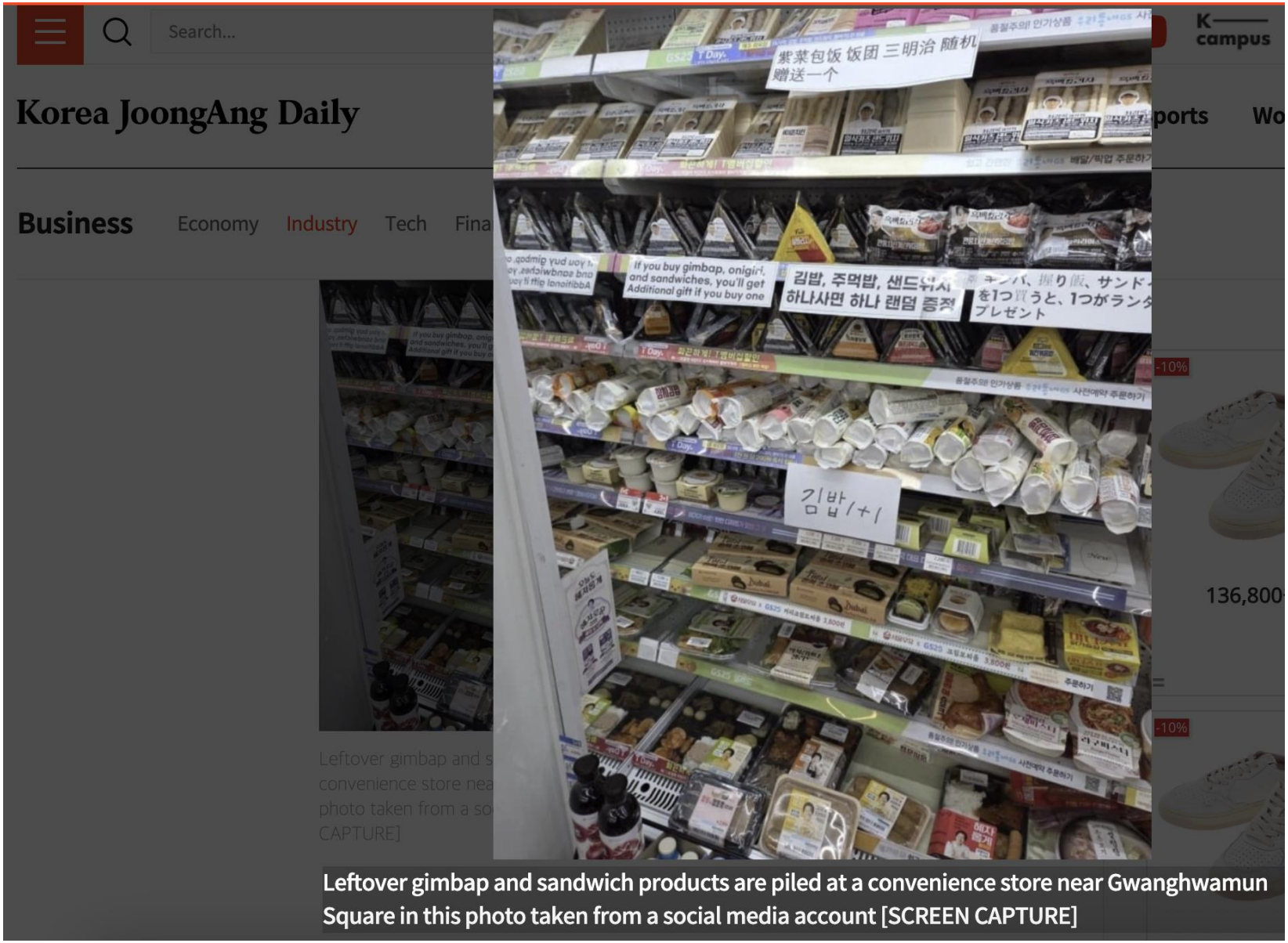
- **SCM systems facilitate efficient customer response**

- ***Push*-based model: Earlier SCM systems**
  - Production schedules are based on forecasts or best guesses of demand.
  - Products are pushed to customers based on these forecasts.
- ***Pull*-based model: Web-based**
  - Actual customer orders trigger events in the supply chain.
  - Transactions are initiated to produce and deliver only what have been ordered.
  - Information flow: retailers – distributors manufacturers – suppliers
- **Internet enables the move from sequential to concurrent SCs**
  - The Internet will ultimately enable a digital logistics nervous system for supply chains, allowing for immediate adjustments and real-time information flow.

Push-Based vs. Pull-Based Supply Chain Models



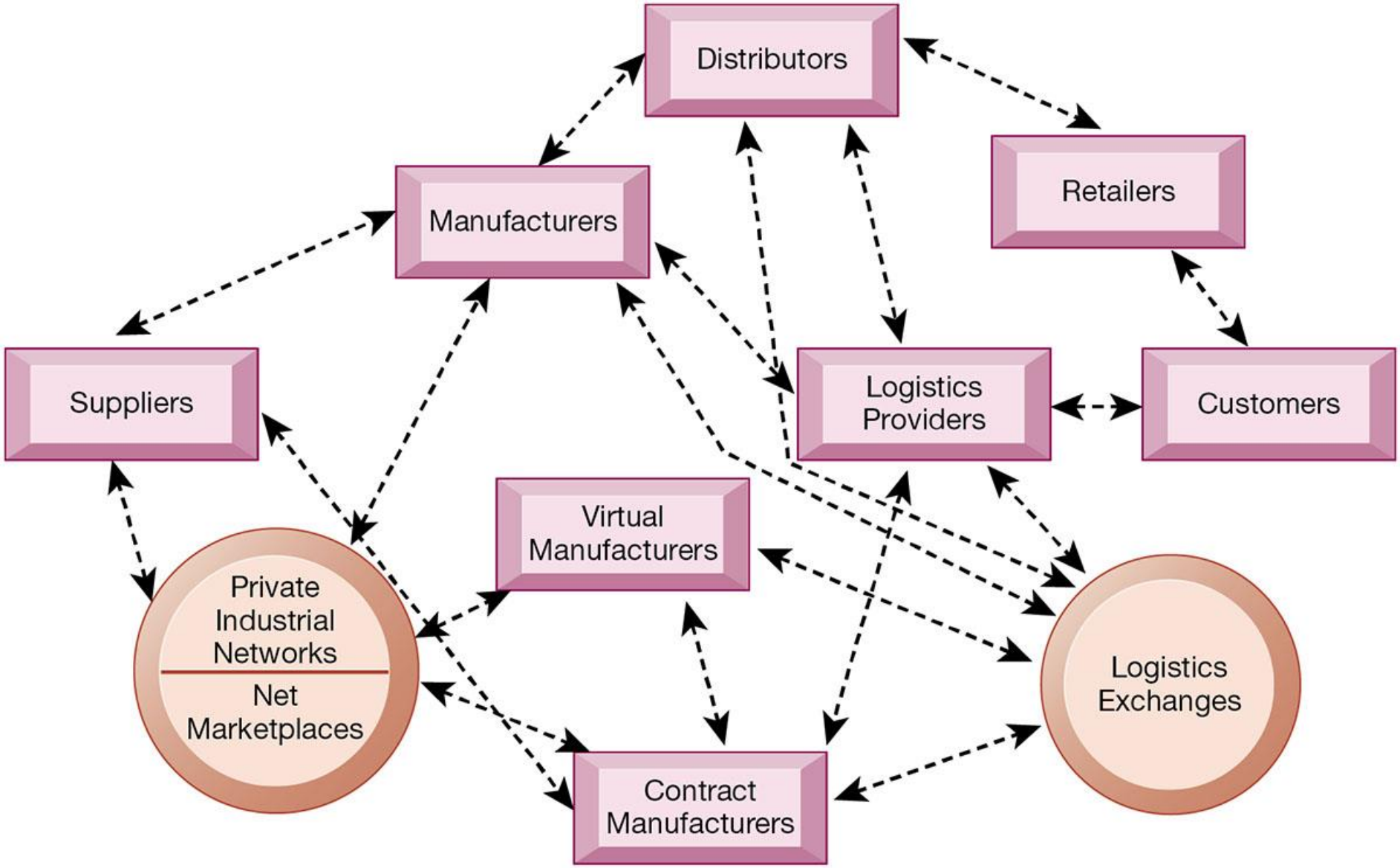
Gimbabs after BTS Concert…



Leftover gimbap and sandwich products are piled at a convenience store near Gwanghwamun Square in this photo taken from a social media account [SCREEN CAPTURE]

https://koreajoongangdaily.joins.com/news/2026-03-22/business/industry/Convenience-store-retailer-sales-boom-during-BTS-comeback-performance/2550771

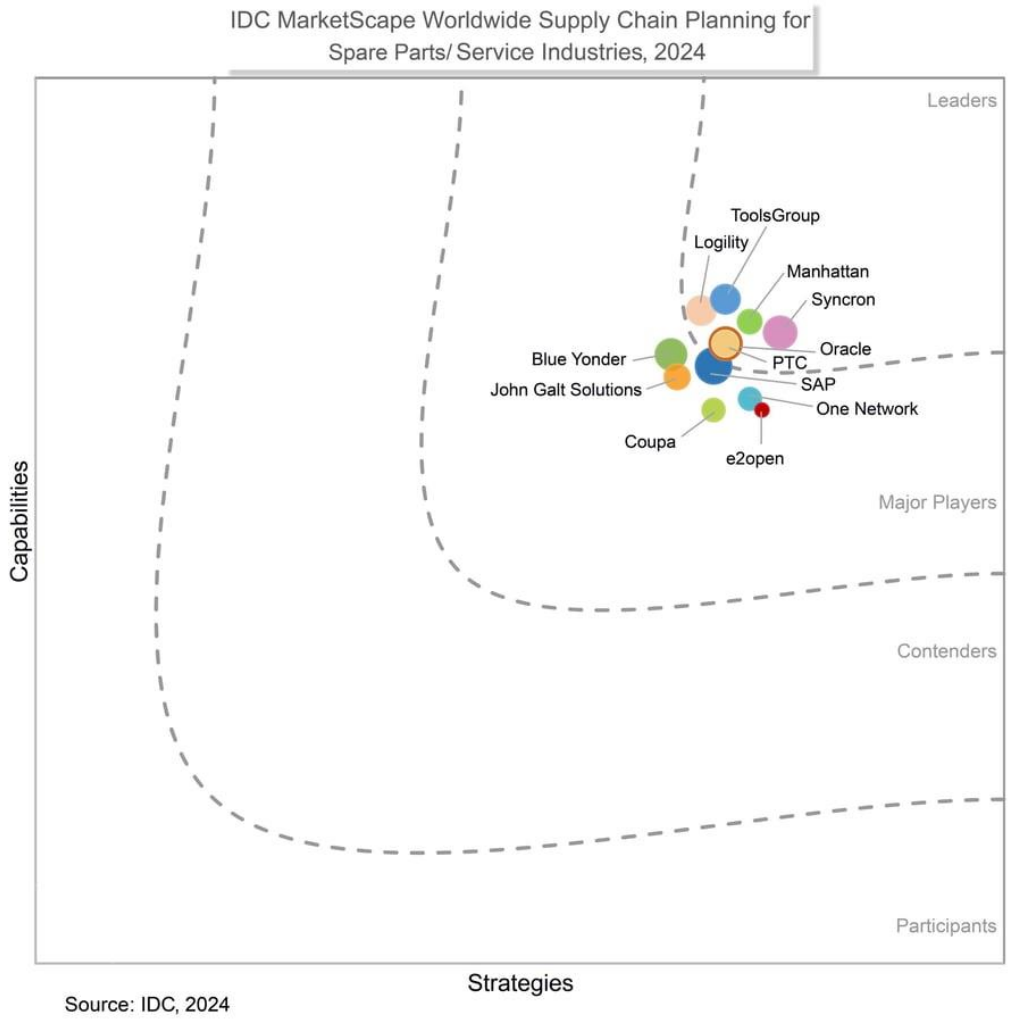
The Emerging Internet-Driven Supply Chain



Business Value of SCM Systems

- **Match supply to demand**
- **Reduce inventory levels**
- **Improve delivery service**
- **Speed product time to market**
- **Use assets more effectively**
  - Total SC costs represent the majority of operating expenses for many businesses and in some industries approach 75% of the total operating budget
- **Increase sales**
  - If a product is not available when a customer wants it, customers often try to purchase it from someone else.

SCM Software



Source: Gartner, IDC

9.3

How do customer relationship management systems help firms achieve customer intimacy?

Customer Relationship Management

- **Knowing the customer:** Who your customers are, how to contact them, whether they are costly to service and sell to, what kinds of products and services they are interested in, and how much money they spend on your company.

- **Make your good customers feel special**

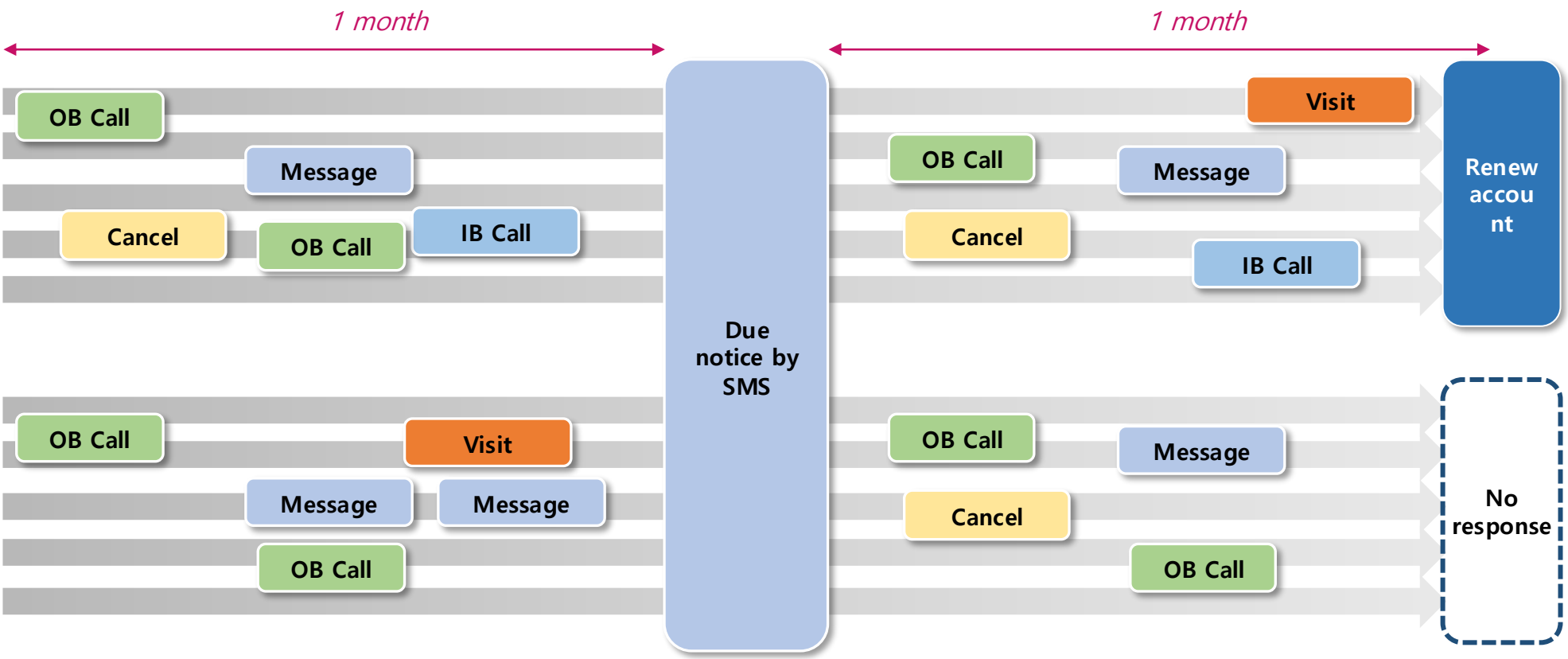
- **In large businesses, too many customers and too many ways customers interact with the firm**

CRM Systems

- **Capture and integrate customer data from across the organization**
  - Consolidate and analyze customer data
- **Distribute customer information to various systems and customer touch points across enterprise**
  - A **touch point** is a method of interaction with the customer, such as telephone, e-mail, customer service desk, conventional mail, SNS, or retail store.
- **Provide data and analytical tools**
  - Helps acquire new customers,
    provide better service and support to existing customers,
    customize their offerings more precisely to customer preferences, and
    provide ongoing value to retain profitable customers.

Example of Touchpoints

- **Example of touchpoints from call centers of a commercial bank**



Customer Relationship Management

Sales Force Automation (SFA)

- **Key Features:**
  - Keeps track of potential and existing customers.
  - Provides details about products and allows customization.
  - Helps generate accurate quotes for customers.
- **Benefits:**
  - ***Personalized Recommendations***: Uses data on past purchases to suggest products tailored to each customer.
  - ***Information Sharing***: Enables sales, marketing, and shipping departments to easily share customer and prospect information.
  - ***Efficiency***: Reduces the cost per sale, and the cost of acquiring new customers and retaining existing ones.
- **Additional Capabilities:**
  - Sales Forecasting. Territory Management, Team Selling

Sales Force Automation Platforms

Source: Gartner, IDC

Marketing

2

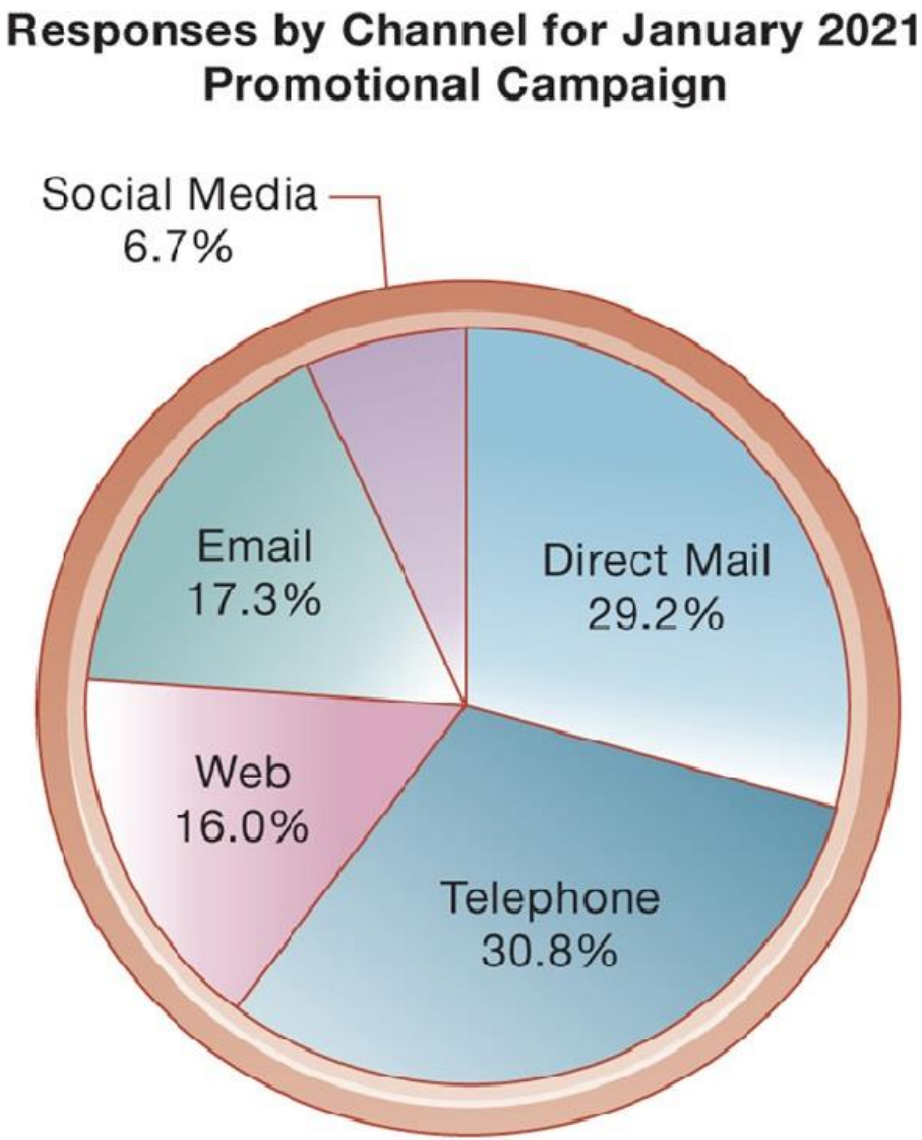
- **CRM systems support direct-marketing campaigns by:**
  - Capturing prospect and customer data
  - Providing product and service information
  - Qualifying leads for targeted marketing
  - Scheduling and tracking direct-marketing mailings or e-mail
- **Analyzing marketing and customer data**
  - Identifying profitable and unprofitable customers
  - Designing products and services to satisfy specific customer needs
- **Managing and executing marketing campaigns at all stages, from planning to determining the rate of success for each campaign.**

Customer Service

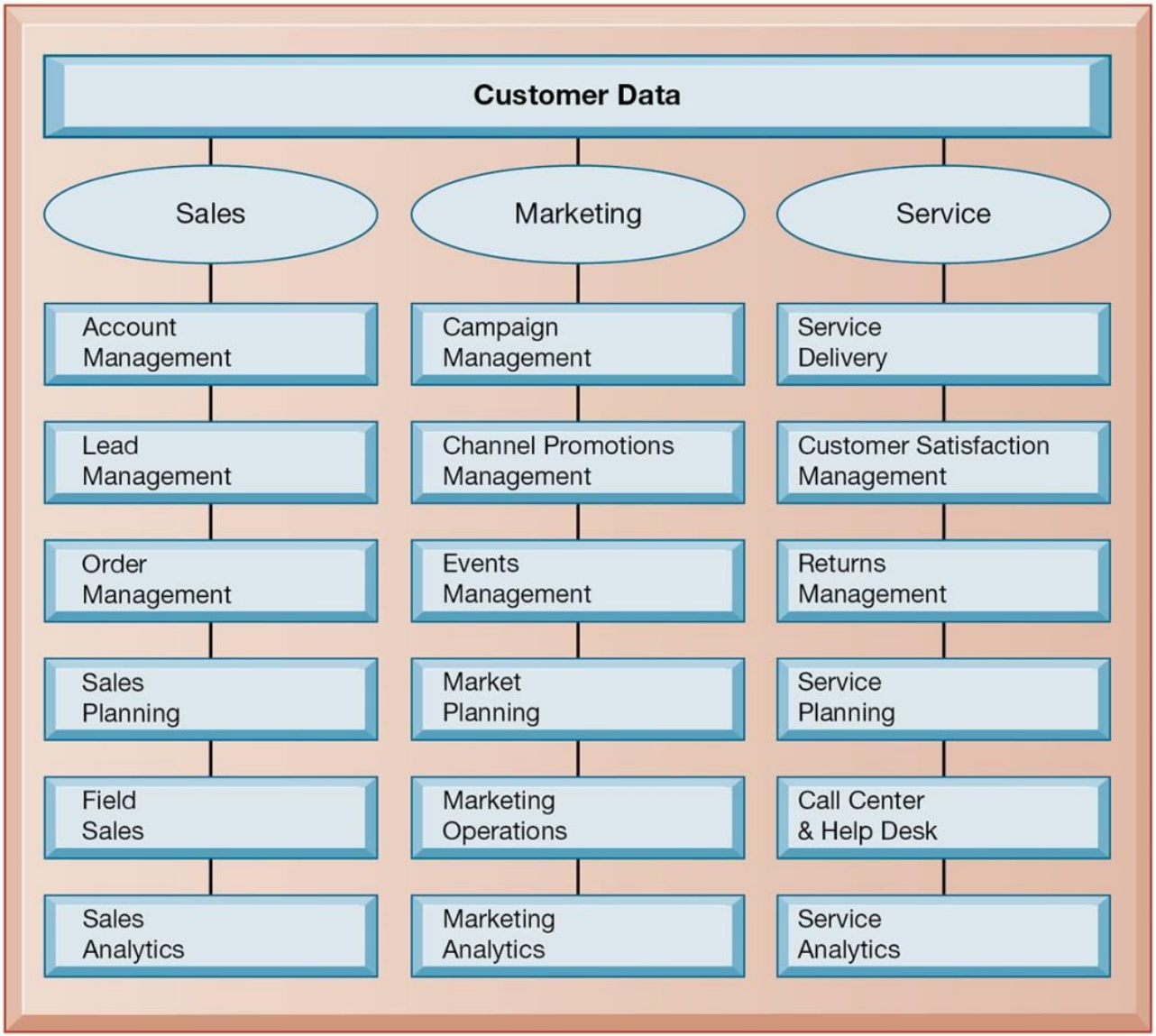
3

- **Providing information and tools to increase the efficiency of call centers, help desks, and customer support staff.**
- **Call Routing and Data Entry**
  - Assign and manage customer service requests efficiently.
  - The service representative inputs the customer's information into the system only once.
- **Consistent and accurate customer information leads to greater productivity, reduced transaction time, and higher quality of service at lower costs.**
  - Customers spend less time on the phone restating their problems, leading to higher satisfaction.
- **Web-Based Self-Service:**
  - CRM systems may include web-based self-service options.
  - Company websites can provide personalized support information and the option for customers to contact service staff by phone for additional help.

How CRM Systems Support Marketing

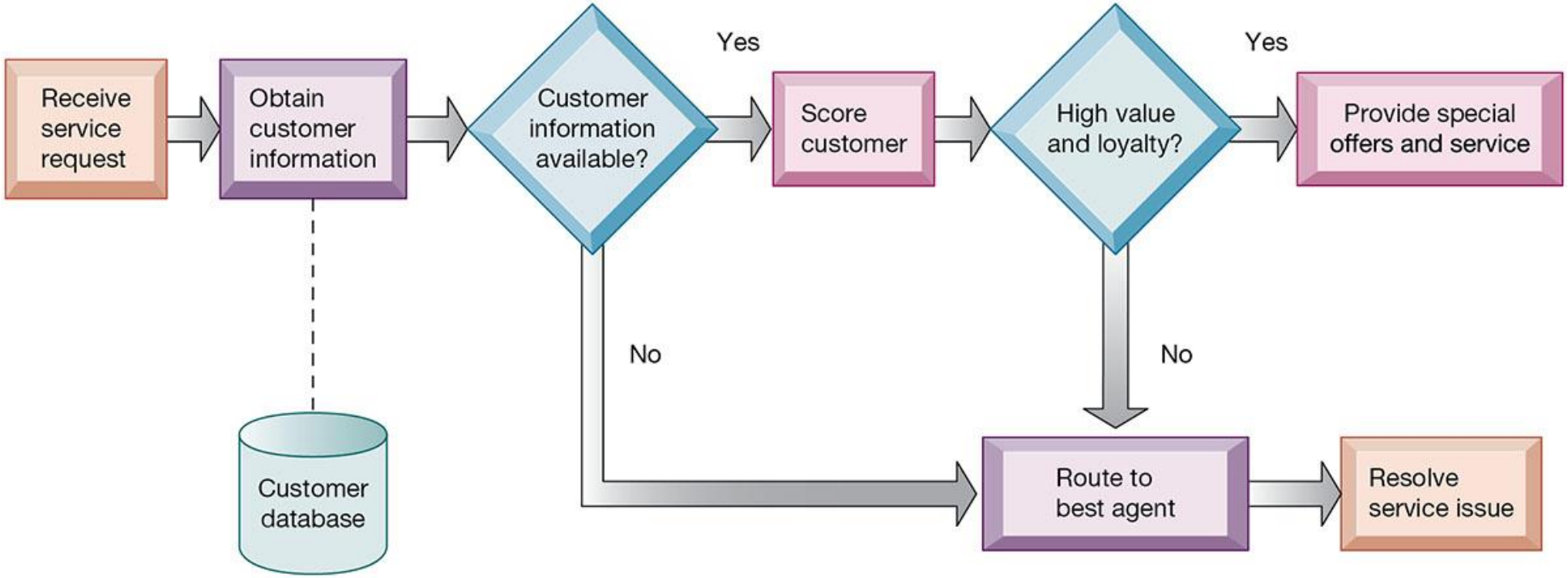


CRM Software Capabilities



CRM Software

- **How a best practice is used for increasing customer loyalty through customer service**
  - Directly servicing customers provides firms with opportunities to increase customer retention by singling out profitable long-term customers for preferential treatment.



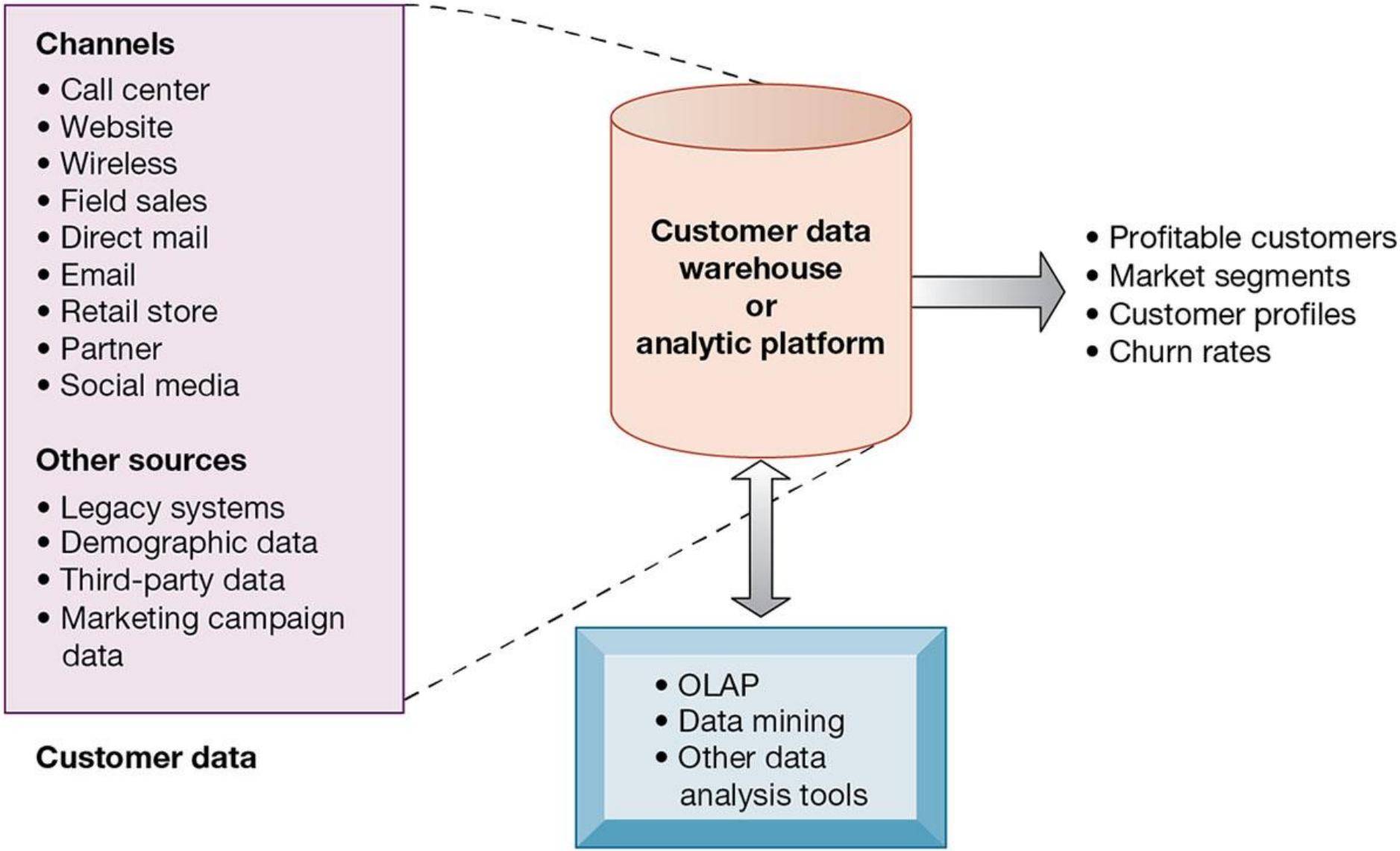
Customer Loyalty Management Process Map

Operational and Analytical CRM

- **Operational CRM**
  - Customer-facing applications
    - Sales force automation, call center and customer service support, marketing automation

- **Analytical CRM**
  - Analyzes customer data generated by operational CRM applications to provide information for improving business performance.
    - Based on **data warehouses** populated by operational CRM systems and touchpoints
    - Based on **other sources** that have been organized in data warehouses or analytic platforms for use in data mining, and other data analysis techniques
    - Customer data can be combined with data from other sources, such as customer lists for direct-marketing campaigns purchased from other companies or demographic data.
  - Data are analyzed to identify buying patterns, to create segments for targeted marketing, and to pinpoint profitable and unprofitable customers

Analytical CRM Data Warehouse



Business Value of CRM Systems

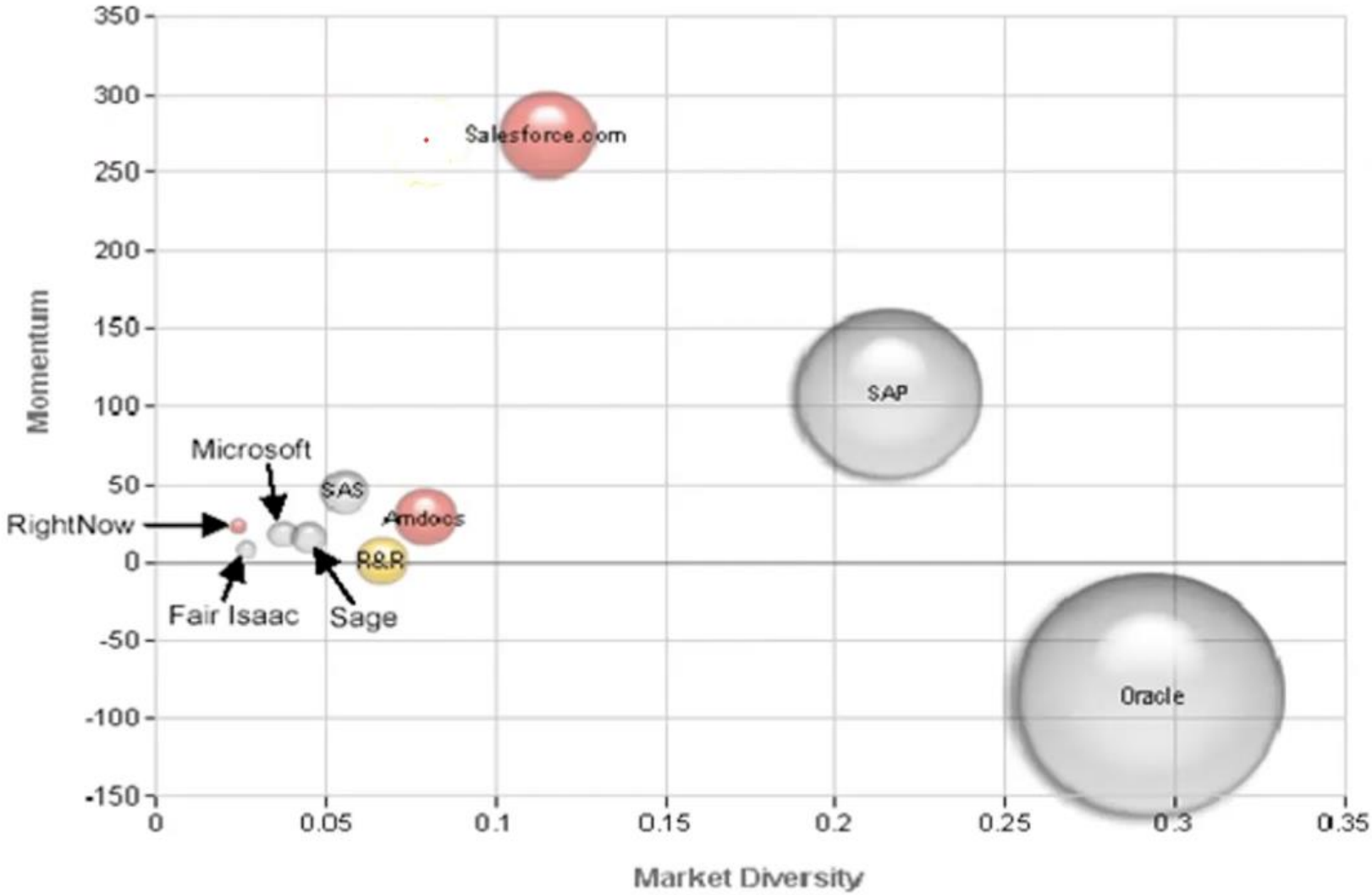
- **Companies with effective CRM systems realize many benefits:**
  - Increased customer satisfaction
  - Reduced direct-marketing costs
  - More effective marketing
  - Lower costs for customer acquisition/retention
  - Increased sales revenue by identifying the most profitable customers and segments for focused marketing and cross-selling

- **Churn rate**
  - Number of customers who stop using or purchasing products or services
  - Indicator of growth or decline of firm's customer base

CRM Software

- **Types of CRM Packages:**
  - Niche tools: Perform specific functions for individual customers.
  - Comprehensive packages: Capture numerous customer interactions, analyze them, and link to other enterprise applications.
- **More comprehensive packages have modules for:**
  - Partner relationship management (PRM): helps a company work better with ***distributors and retailers*** by sharing information, managing leads, pricing, promotions, orders, availability, and partner performance.
  - Employee relationship management (ERM): Manages ***employee-related issues linked to CRM***, includes setting objectives, performance management, performance-based compensation, and training.
- **Major CRM application software vendors include Oracle, SAP, Salesforce.com, and Microsoft Dynamics CRM.**
  - Provide software and online tools for sales, customer service, and marketing.

CRM Software



자료: IDC Competitive Analysis report, "Worldwide CRM Applications 2006 Vendor Shares: Marketing, Sales, and Customer Service"

9.4

What are the challenges that enterprise applications pose, and how are enterprise applications taking advantage of new technologies?

Enterprise Application Challenges

- **High Cost and Complexity**
  - Enterprise applications are ***expensive*** and difficult to implement.
  - Heavy customization often increases both ***time and cost.***
- **Organizational and Operational Change**
  - Implementation requires changes in business processes, roles, and work routines.
  - Employees must learn and understand how their data affects other functions.
- **Data, Integration, and Vendor Dependence**
  - ERP, SCM, and CRM require standardized, accurate, and well-organized data.
  - Relying on one vendor can create high switching costs and long-term dependence.
- **How Companies Reduce These Challenges**
  - Use simplified or cloud-based solutions for flexibility.
  - Minimize unnecessary customization and adopt best practices where possible.
  - Align business processes to the software rather than over-modifying the software.

Next-Generation Enterprise Applications

- **Enhanced Flexibility and Integration:**
  - Becomes more flexible, user-friendly, web-enabled, and mobile-compatible.
  - Stand-alone systems are becoming obsolete, replaced by integrated enterprise solutions or e-business suites ➔ ensuring different systems ***work together*** and ***connect*** with customer and supplier systems.
- **Next-Generation Features:**
  - Cloud solutions and mobile functionality.
    - More companies are running all or part of their enterprise applications in the cloud.
    - Major vendors provide versions of their products that work on mobile devices
  - Enhanced CRM with social networking technologies
    - Leading CRM vendors offer tools to monitor, track, and analyze social media activity on platforms such as Facebook, LinkedIn, Twitter, and YouTube.
  - Incorporating AI (e.g., Salesforce.com's Einstein AI tool)

Key Takeaways

- Enterprise systems integrate the whole firm through connected software modules and a shared central database, so information entered in one process becomes available immediately across other functions.
- They promote standardization and centralization by enforcing common data standards, business processes, and a unified technology platform across the organization.
- SCM systems improve supply chain decisions by automating information flows across partners, helping firms decide what to buy, make, and ship while reducing uncertainty and the bullwhip effect.
- CRM systems create a company-wide view of customers by integrating sales, marketing, and service, which helps firms improve service, identify profitable customers, and reduce churn.
- Successful implementation is difficult and costly because enterprise applications require major organizational change, employee training, process redesign, and strong data management.
- Modern enterprise applications are becoming more flexible through web-based integration, cloud and mobile deployment, open-source or on-demand options, and added social networking capabilities.

Next Week…

- **Next week, we will cover:**
  - [Chapter 4] Ethical and Social Issues in Information Systems

Q & A